

MATH0053 Algebra 4: Groups and Rings

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0053
<i>Level:</i>	6 (UG)
<i>Normal student group(s):</i>	UG: Y2 Mathematics programmes
<i>Value:</i>	15 credits (7.5 ECTS credits)
<i>Term:</i>	2
<i>Assessment:</i>	90% examination, 10% coursework
<i>Normal Pre-requisites:</i>	MATH0006 (Algebra 2)
<i>Lecturer:</i>	Dr N Kalaydzhieva

Course Description and Objectives

This module builds on the notions of groups and rings covered in Algebra 2. It is divided into two parts in the approximate ratio 3:2 (groups : rings).

In part 1 (groups), we develop a strong foundation of the theory of finite groups. Students will become familiar with the structure and classification of groups of small order, with emphasis on understanding the methods used to classify finite groups. These ideas are developed progressively, culminating in the study and application of Sylow's Theorems.

In part 2 (rings), we introduce ring theory, extending familiar arithmetic to more general algebraic systems. Topics include ideals, quotient rings, polynomial rings, and principal ideal domains. The module concludes with the construction of finite fields using quotients of polynomial rings.

Recommended Texts

- JB Fraleigh, *A First Course in Abstract Algebra* (Pearson)

Detailed Syllabus

- Recap: groups, subgroups, cyclic, dihedral, symmetric, matrix groups, cosets, Lagrange's theorem.
- Cauchy's theorem, normal subgroups and quotient groups.
- Group homomorphisms: homomorphisms, isomorphisms, kernels and images, the first isomorphism theorem, automorphisms, and the automorphism group of a group.
- Direct, semi-direct products and recognition criteria.
- Group actions: orbits and stabilisers, the Orbit–Stabiliser theorem, conjugacy classes, the class equation, applications to counting and group structure.
- Sylow theory: Sylow p -subgroups, Sylow theorems, applications to the classification of groups of small order.
- Rings: examples of rings, integral domains and fields, ring homomorphisms, ideals, quotient rings, maximal, prime ideals, and principal ideal domains.

- Polynomial rings: polynomial arithmetic, irreducible polynomials, factorisation of cyclotomic polynomials, construction of finite fields as quotients of polynomial rings.

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